

Research Profile

Final-year PhD researcher in systems neuroscience with experience in auditory neuroscience, in vivo mouse electrophysiology, two-photon calcium imaging in awake mice, optogenetics, stereotaxic viral injections, behavioral task design, EEG, and neural data analysis. My thesis work combines end-to-end experimental setup development, data acquisition, MATLAB-based analysis pipelines, and manuscript preparation. I currently have two manuscripts under review from my in vivo electrophysiology and two-photon imaging work.

Education

- 2021–
Ongoing **Research Scholar**, *Advanced Technology Development Center, IIT Kharagpur.*
- 2012–2014 **Master of Technology in Chemical Process Control**, *University of Calicut.*
GPA: 8.09/10
- 2004–2008 **Bachelor of Technology in Electronics and Instrumentation**, *MG University.*
GPA: 77.78/100

Research Projects

- 2024–
Ongoing **Processing of regular and irregular sound streams using EEG in humans**, *Information Processing Laboratory, IIT Kharagpur, PI: Dr. Sharba Bandyopadhyay.*
Designed active/passive auditory paradigms; built an EMOTIV-EEG workflow from task design and data collection to preprocessing and analysis in MATLAB. Manuscript in preparation
- 2021–
Ongoing **Processing of regular and irregular sound streams in primary and secondary auditory cortex**, *Information Processing Laboratory, IIT Kharagpur, PI: Dr. Sharba Bandyopadhyay.*
Performed in vivo electrophysiology and two-photon calcium imaging in awake mouse auditory cortex, including A1 and secondary auditory regions (AUD/AUv); characterized response dynamics, adaptation, and selectivity across regular and irregular sound sequences; developed reproducible MATLAB analysis pipelines. This work has resulted in two manuscripts currently under review.
- 2021–2023 **Auditory discrimination behavior rig development and mouse training**, *Information Processing Laboratory, IIT Kharagpur, PI: Dr. Sharba Bandyopadhyay.*
Built and integrated a complete auditory go/no-go mouse behavior training setup from scratch, including IR nose-poke sensors, lickometer, solenoid-based water reward delivery, and aversive foot-shock feedback; developed adjustable stimulus-discrimination protocols and implemented MATLAB/NI-DAQ-based task control.

Technical Skills

- In vivo systems neuroscience** Awake mouse electrophysiology; Two-photon calcium imaging; Optogenetics; Stereotaxic viral injections using Nanoject.
- Neural data analysis** MATLAB pipelines for single-neuron, LFP, calcium imaging, and EEG analysis; EEG preprocessing/cleaning in EEGLAB; stimulus-response and selectivity analysis; functional connectivity/cross-correlation analyses; statistical testing and visualization.
- Programming & tools** MATLAB; LabVIEW; Arduino; NI-DAQ workflows; Git/GitHub; C/C++.

Publications and Presentations

Neuroscience

Micheal, A. S. M., Bandyopadhyay, S. (2026). "Inter-token silence period spiking activity enhances selectivity of distinct groups of auditory cortical neurons to periodic and aperiodic sound sequences." *bioRxiv*. <https://doi.org/10.64898/2026.02.03.703433v1>

Micheal, A., Bandyopadhyay, S. "Processing of regular and irregular sound streams in mouse auditory cortex." *Society for Neuroscience*, 2024, PST403.

Micheal, A. S. M., De, A., Bandyopadhyay, S. "Selectivity to periodic sequences of sound in auditory cortical single neurons." *IBRO Neuroscience Reports*, 15, S695.

De, A., Raghavendra, K., **Soniya, A.**, and Bandyopadhyay, S. "Differential Effects of Two-Tone Harmonics and Single Tone Tokens on Subsequent Sounds." *Association for Research in Otolaryngology*, 2023, Orlando, FL.

Earlier Engineering Publications

Ann Soniya M. Michael, Manoj, N. "Design of Fuzzy Logic Controller for Continuous Crystallization." *International Conference on Advances in Chemical Engineering and Technology*. Publisher: Elsevier, ISBN: 9789351072843.

Marymol, P., **Ann Soniya M.**, Reshma, K. V., Reema, M. "Automatic Fire Fighting Robot." *International Journal of Advanced Research in Computer Science and Software Engineering*, Vol. 5, Issue 5, May 2015.

Ann Soniya, Keerthi Chacko, Avinash K. K., Akhil Jose. "Implementation of Solar Powered Voice Recognition Robot." *Global Journal for Research Analysis*, Vol. 4, Issue 5, May 2015.

Teenu Jose, Divya K. V., Marymol Paul, **Ann Soniya**. "A Method for Reducing Three Phase Power Capacitor Switching Transients." *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, Vol. 4, Issue 3, March 2015.

Teaching and Mentoring

2022–2025 **Teaching Assistant, NPTEL.**

Teaching assistant for *Cognition and Computation* and *Computational Neuroscience*.

2014–2021 **Assistant Professor, Vimal Jyothi Engineering College and Sahridaya College of Engineering and Technology.**

Taught biomedical engineering, instrumentation, process control, and control theory courses; coordinated the Biomedical Engineering program at Sahridaya; supervised undergraduate and graduate biomedical instrumentation projects.

Earlier Engineering and Industry Experience

2017–2020 **Biomedical instrumentation and assistive technology projects, Sahridaya College of Engineering and Technology.**

Led and supervised biomedical device projects involving EEG/EMG-based assistive systems, rehabilitation devices, eye-gaze communication aids, virtual-environment-based training tools, and Arduino/PLC-based medical-device prototypes.

2013–2014 **Research Project Intern, APEP, Indian Space Research Organisation.**

Master's thesis project: MATLAB/Simulink-based modeling and fuzzy logic controller design for a continuous crystallization unit.

2009–2012 **Test Engineer, Infosys Technologies Limited.**

Software testing for point-of-sale systems.

References

Dr. Sharba Bandyopadhyay – Assistant Professor, ECE, IIT Kharagpur; sharba.ban@gmail.com

Dr. Nihar Ranjan Jana – Professor, Dept. of Bioscience, IIT Kharagpur; nihar@iitkgp.ac.in